Strata — Developer Guide

For engineers implementing tickets

strata-kb v1.18.0 · 2026-09-19

1. Who this guide is for

You write product code. Tickets arrive already grounded in the organisation's knowledge base, with citations pinned to a specific version of a specific document, and you implement them.

Strata does two things for you.

It tells you what the source actually said. Not "check the handbook" — the exact section, at the exact revision the analyst was reading, with a warning if it has changed since.

It publishes what your codebase is, so everyone else — analysts drawing architecture diagrams, other teams, future you — can search your services, dependencies, endpoints and schema without reading your repository.

1.1 What a dev repo does and does not do

Does	Does not
Read the hub's published knowledge, including the BA's published tickets	Ingest documents from outside the repo
Publish knowledge about **its own code**	Publish domain knowledge
Run a five-command implementation workflow	Merge or ship anything by itself

The boundary is deliberate. A `child` repository brings in outside documents; a `dev` repository ingests nothing and authors only knowledge about itself. That is what makes it safe for your CI to publish continuously.

1.2 The two documents your repo publishes

Document	Origin	Regenerated	Needs review
`<repo_id>-code`	`kb code-ingest` — deterministic, no model	on demand, from the working tree — never committed; CI publishes it on every push to the default branch	no
`<repo_id>-svc`	drafted by a model from that evidence, corrected by you	never	always

`-code` is machine fact: service names, technologies, dependencies, commands, endpoints, schema, folder structure. `-svc` is human claim: what each service is actually *responsible for*.

They are joined by section id — `svc.<name>` exists in both — so an analyst drawing a C4 container gets the alias, label and technology from `-code` and the description from `-svc`.

2. Setup, once

```
pip install strata-kb
kb init --kind dev
```

2.1 Configure

```
# .kb/config.yaml
kind: dev
repo_id: payments
hub: https://github.com/acme/kb-hub.git
intake: https://kb-intake.acme.com
```

- **`hub`** is the only read source for `kb query`, `kb get`, `kb resolve` and MCP.
- **`intake`** is how CI publishes. Ask the hub maintainer to add this repository under `repos:` in the hub's `federation/registry.yaml` — publishing is refused until they do.

If the KB should carry SQLite schema, or extra index tags, say so here — not in `kb-code.yml` — so every local refresh and CI ingest the same way:

```
code_ingest:
 db: [data/app.sqlite]   # explicit only
 tags: [payments]
```

2.2 Connect your assistant

`.mcp.json` (Claude Code) and `.cursor/mcp.json` (Cursor) were scaffolded already wired to the hub's MCP tools, but through placeholders — `${STRATA_KB_HUB_URL}` and `${STRATA_KB_HTTP_TOKEN}` — that nothing has set yet.

Run `kb mcp-setup` (in your assistant: `/kb-mcp-setup`). It asks for the hub's HTTP base URL and token, writes both into `.env`, makes sure git ignores that file, and then verifies them against the hub, so a wrong URL and a rejected token come back as different errors:

Variable	Example
`STRATA_KB_HUB_URL`	`http://kb-hub.example.com:8321`
`STRATA_KB_HTTP_TOKEN`	the hub's bearer token

A bare re-run verifies again without retyping the token. Prefer the hidden prompt or the `STRATA_KB_HTTP_TOKEN` environment variable over `--token` — that flag leaves the token in your shell history.

`.mcp.json` and `.cursor/mcp.json` read those two variables from your environment, not from `.env` directly, so load `.env` into your shell (`set -a; source .env; set +a`, or use direnv) and restart your assistant — MCP reads the environment only at startup.

2.3 Turn on the PR gate

`kb init` writes `.github/workflows/kb-pr-lint.yml` but cannot enable branch protection for you. Add **`pr-lint` to the branch's required checks**.

> Decide this knowingly. Unlike the BA repo's gate, this one never self-skips. Once required, it blocks **every** pull request lacking the required sections — including bot pull requests such as a dependency bump or a revert.

2.4 Ask about the hub's auto-merge policy

`-code` pull requests on the hub may be auto-merged; `-svc` pull requests must never be. Ask your hub maintainer how it is configured, and make sure of one thing:

> `kb ci-publish` opens **one** hub pull request per push, not one per document. The intake keys the publish branch by repository id and reuses it while a PR is pending — so a push carrying a regenerated `-code` alongside an already-merged `-svc` amend lands **both** in that same PR.
>
> The auto-merge rule must therefore be **path-scoped to `.kb/<repo_id>-code/**`**, never a whole-PR or whole-repo rule. A rule that fires on "this is a dev repo's `-code` publish" silently defeats the `-svc` review gate the two-document split exists to protect.

This auto-merge rule is also what keeps the hub from lagging main: without it, every push adds to one pending hub PR that someone has to merge. `kb doctor` reports when the hub's `-code` lags.

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3. Onboarding an existing project

Your repository already has services doing real work. Adopting Strata should not mean documenting the system from a blank page, so there is a one-time seed.

```
/dev-code-seed
```

Seven steps:

#	Step	Who
1	**Preflight** — confirms `hub:`, `repo_id:` and `intake:` are set and the repo is allowlisted on the hub; warns on a dirty tree	agent
2	**Extract** — `kb code-ingest --scaffold-svc` regenerates `-code` and creates a `pending` `svc.<name>` scaffold per detected service, backed by deterministic code evidence	agent
3	**Draft** — `kb summarize <repo_id>-svc` writes a responsibility paragraph for each from that evidence	agent
4	**Review** — walk every drafted section against its evidence and correct it	**you**
5	**Approve** — `kb approve <repo_id>-svc` flips corrected sections to `reviewed`	you
6	**Flows** (optional) — add `flow.<name>` sections for business flows crossing several services	you
7	**Validate and publish** — `kb build` without `--allow-pending`, then `kb publish --pr`	you

3.1 Step 4 is the actual work

> **The model's draft is a draft, not a fact.** It can name the wrong flow, miss a responsibility, or over-claim one. Approving a section you have not read defeats the entire point of the gate.

Read each `svc.*` section's summary next to its code evidence and correct it. Where a responsibility touches a domain standard, cite `doc-id §section` from the hub instead of restating the rule in your own words.

Budget 10–15 minutes per service. That is the honest cost, and it is a one-time cost. After the seed, `-code` regenerates itself and `-svc` accrues a few lines at a time.

For step 6, skip freely. A missing flow beats a guessed one.

3.2 Why the first publish is `--pr` by hand

The seed happens *before* anything has merged to the default branch, so there is no push for `kb-code.yml` to react to.

Use `kb publish --pr`. Plain `kb publish` would tag the commit and poll the intake for up to ten minutes waiting on a tag-triggered CI run this repository does not have, then fail with a misleading error.

That hub pull request is reviewed by an analyst or architect, like any other hub content. `-svc` is curated knowledge; it publishes through review, never auto-merge.

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4. Implementing a ticket

Five commands, scaffolded for Claude Code, GitHub Copilot and Cursor.

```
/dev-implement-ticket <ticket>     intake · check · resolve · ground · placeholders
        │
        ├─▶ dev-design        ── A1: independent reviewer ── GATE 1: you approve the design
        ├─▶ dev-plan          ── A2: independent reviewer ── GATE 2: you approve the plan
        ├─▶ dev-execute          A3 per task · A4 at branch end  (repeatable, resumable)
        └─▶ dev-handover      ── A5: merge-risk review ── GATE 3: you open the PR
                                 GATE 4: you merge it
```

`/dev-implement-ticket` is also the orchestrator: given a fresh ticket it runs the first five steps, then invokes the phases in order, detecting how far a ticket already got so it never redoes finished work.

4.1 A ticket id is enough

Give the assistant only a ticket id (`M-<slug>-US<n>`, or a plain ticket stem). Intake gets the ticket **from the hub first**: the BA publishes every ready ticket in the `<repo_id>-plan` document, so the hub is the source of truth.

```
kb ticket fetch M-checkout-US3          # or <repo>:<id> if several repos hold it
```

- It finds the one federation repo whose `*-plan` document has that section. If none does, it exits `1`: the ticket is not published yet, so ask the BA to `kb publish`, or pass a path or paste the body. If several do, it exits `1` and names them; qualify the id as `<repo>:<id>`.
- It writes `docs/impl/<id>/ticket.md` (`--out` to change it) with the title, one provenance line and the ticket body: `> Fetched from hub: <repo>:<doc> §<id> @ <hub rev> — do not edit; amend in the BA repo.`

- It **overwrites** the file on every fetch. The copy is derivable from the hub, so `docs/impl/.gitignore` ignores `*/ticket.md`: never commit it, and never edit it. If the ticket is wrong, the BA amends it in the BA repo.
- An id that is not a plain name (a path, `..`) is refused.

Without the CLI, the assistant reads the section through MCP (`kb_get_section`, `<doc> §<id>` at `l3`) and saves it the same way. A path or a pasted ticket body still works, as the fallback when the hub has no copy or cannot be reached.

The hub copy has **no** `## Definition of Ready`: being on the hub means the BA ticked every box, so no step may stop on its absence.

The parent mission's `## Sequencing` is read the same way when there is no local mission file (`kb get <doc> <M-id> --level l3`), so you can see what the ticket is blocked by without cloning the BA repo.

The grounding check

Right after Intake the flow runs `kb ticket check docs/impl/<id>/ticket.md` (your `<ticket-file>`; skipped with a note when the ticket has no `## Technical grounding`). It reads the hub's `-code` and, for a `[NEW: D<n>]`, the parent mission from the hub's plan when no local `missions/` holds it. The `Grounded on:` revision moves with every commit you merge, so a mismatch alone never fails:

Output	Meaning	What to do
PASS, `[note] grounded @<old>, still valid at @<new> — cited sections unchanged`	Nothing the ticket cites changed	Carry on
PASS, `[warn] … §<id> changed; cited names still hold, review the diff:`	A cited section changed, every cited name is still there	Read the diff and decide; carry on unless it changes the ticket's meaning
PASS, `[warn] some cited facts could not be verified …`	A check was skipped (e.g. a tree cap)	Check those facts by hand
FAIL, `… re-run /sa-ticket-ground`	A cited id, column, route, command, file or compose fact is gone	Stop: a blocker for the BA, whose SA re-grounds the ticket
FAIL, plus `[note] checked against the local -code, not the hub's copy …`	The hub was unreachable or lacks `-code`; your branch may be the cause	Re-run once the hub is reachable and holds the `-code`
FAIL without the re-ground line	An open decision or an undecided `[NEW: D<k>]`	Report it to the BA and stop

What a ticket's status means to you

The hub's **Plans** menu (`/ui/plans`) and `kb mission next --from-hub [--repo <ba-repo>]` show every story of a published mission as **done** (its id is in a `hist.*` row of your `-svc`, which `kb svc note` writes at handover), **ready** (its ticket is published), **not ready** (no published ticket yet) or **blocked** (with reasons). Pick a **ready** one.

On the BA side, `kb mission next` can also report a ticket `draft` with the reason `parent mission rev <M> affects this story`: the mission was amended after the ticket was written, and the BA has not re-authored it. Such a ticket is not published, so you see the story as *not ready* and `kb ticket fetch` finds nothing. Wait for the BA; do not implement from an older copy.

4.2 Pick your entry point



Every entry point re-checks citation freshness first. The hub may have published since your last session, so a reference that was `ok` yesterday can be `stale` today. Checking only at handover would be too late.

4.3 Where work lives

File	Written by	Purpose
`docs/impl/<id>/design.md`	`dev-design`	the design
`docs/impl/<id>/plan.md`	`dev-plan`	one task per acceptance criterion, `- [ ]` checkboxes, a `Depends on:` line per task
`docs/impl/<id>/ticket.md`	`kb ticket fetch`	the ticket fetched from the hub; gitignored, overwritten on every fetch, never edited
`docs/impl/<id>/context.md`	`dev-implement-ticket`	resolved-context cache; gitignored, regenerated on demand
`docs/impl/<id>/review/`	`dev-execute`, `dev-handover`	per-task diffs and reviewer reports, the branch diff, the merge-risk report; gitignored, derivable from git and the review records below

`kb init --kind dev` and `kb impl tidy` move an old flat layout (`docs/impl/<id>-plan.md`, …) into these folders.

The design and plan files each grow a `## Review record` table as their reviews run — the committed history of every round, not a separate file.

State is derived from the first two files, the current branch, and whether a PR is open — **never from conversation history**. Any phase resumes cold in a brand new session.

`kb init` never edits your `docs/impl/` content: it adds `.gitkeep` and a `.gitignore` so the context cache, the fetched ticket and the review artefacts never land in a pull request, and moves an old flat layout into `<id>/` — never overwriting. Commit the renames.

Waves and lanes. Every task in the plan carries a `Depends on:` line under its `### Task <n>:` heading — `none`, or the tasks whose Interfaces it consumes or whose files it shares. `dev-plan` derives it; the A2 reviewer checks it. `kb plan waves docs/impl/<id>/plan.md` turns those lines into waves (wave n = tasks whose dependencies all sit in earlier waves) and fails when two tasks share a file without a dependency between them, when there is a cycle, or when a line is missing — a plan like that has no safe parallel order. A plan where every task lacks the line predates waves and runs sequentially.

`dev-execute` runs it after isolating the branch. A wave of one task runs as before; a wave of several runs each task in its own git worktree lane cut from the ticket branch, at most 3 lanes at a time, merged back in task order with `--no-ff`, the full suite run once per wave. A merge conflict means the plan was wrong and goes back to `dev-plan`.

4.4 Four human gates, five agent review rounds

Nothing in this pipeline merges or ships without a human.

1. **Design approved** — you sign off before any plan is written.
2. **Plan approved** — you sign off before any code is written.
3. **PR opened** — the agent writes the body; you open it.
4. **Merged** — you review and merge.

The agent does neither of the last two itself.

Before each of the first three gates, a different agent has already read the work back in a context of its own — never the same agent that produced it:

	Where	What it looks at
A1	`dev-design`, before GATE 1	whether the design matches the ticket
A2	`dev-plan`, before GATE 2	whether the plan is test-first and runnable
A3	`dev-execute`, per task	whether this task's diff meets its spec, and whether the code is good
A4	`dev-execute`, after the last task	whether the finished branch rounds out the ticket
A5	`dev-handover`, before GATE 3	whether merging into the default branch is safe

Criteria live in `docs/pr-review-rubric.md`, overridden by `docs/pr-review-rubric.local.md`. Each round allows at most 3 fix/review passes. A round only counts as clean once no BLOCKER and no SUGGESTED finding is left open — NOTE and NITS get recorded, not fixed — and a BLOCKER or SUGGESTED still standing after round 3 stops the flow and hands it to you.

Where your assistant's runtime cannot dispatch subagents — the default for the Copilot and Cursor wrappers — the review still runs, but as a single pass reading only what it was handed, and it is a **report, not a gate**: the phase stops there and hands the result to you instead of advancing on its own say-so.

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5. What is enforced, and what is only asked

Be clear-eyed about the difference.

5.1 Machine-enforced

Rule	Enforced by
The PR carries its evidence	`kb pr lint` in `kb-pr-lint.yml` on every pull request
Unreviewed knowledge cannot reach the hub	`kb build` exits 1 while any `-svc` section is `pending`
The context cache is CLI-owned	`kb resolve --status-only --cache` refuses a cache whose version, ref set or resolved block differs from the ticket's
The merge-risk review has run and no BLOCKER remains	`kb pr lint` — the `## Testing` section must carry a `Blocking: No` line

The PR body is short: `## Ticket` (what was asked), `## Changes` (what this PR did), `## Testing` (AC→test line, the pasted test command with its summary lines, A5's `Blocking: No`), and an optional `## Notes` (out of scope, not handled, findings). `kb pr lint` fails when Ticket, Changes or Testing is missing or still holds the template's comment, when Testing has no pasted output or no fence containing the plan's test command, or when its `Blocking:` line is missing or reads `Yes`.

5.2 Prompt-only — stated in every wrapper, measured by nothing

These are rules the skills repeat. Nothing in `kb` enforces or measures them. The only check is the human at the gate — which means you.

- **TDD.** No production code without a failing test observed first, at every step of `dev-execute`. Exempt classes are named in `docs/tdd-exemptions.md` and declared in the plan.
- **Shown verification.** No completion claim without the real command output. `kb pr lint` checks the PR body, not the session.
- **Pinned values, verbatim.** Every standard-derived value comes from the resolved section at its pinned hub version, with a citation comment.
- **The ticket is read-only.** Findings go back to the analyst; the agent never edits the ticket.
- **Gates 1 and 2.** Design and plan approved before the next phase. The `status:` header records it; a human grants it.
- **`OPEN(BA)`.** An ambiguous acceptance criterion is escalated, never reinterpreted.
- **A1–A4.** The four agent reviews before GATE 3 are prompt discipline. Nothing in `kb` measures whether they ran. Only A5 has teeth: `kb pr lint` reads the `Blocking:` line in the PR's `## Testing` section. An agent that skips A1–A4 and writes an honest `Blocking: No` still passes — the gate sits where the merge actually happens, not at every step before it.

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6. Working with citations

```
kb resolve tickets/ABC-123.md    # ok | stale | broken, per ref
kb diff hr-handbook --against <rev>
kb get hr-handbook 4.12 --level l3
```

Status	Meaning	What to do
`ok`	The pinned content still matches what is published	Implement against it
`stale`	The source was amended after the ticket was written	Run `kb diff`, then ask the analyst whether the acceptance criteria still hold. **Do not silently implement the new version.**
`broken`	The reference no longer resolves	Escalate. The ticket is not implementable as written.

When a value in your code comes from a standard — a format, an enum, a threshold, a field length — take it from the resolved section at the pinned version and leave a citation comment next to it. Six months later that comment is the only thing connecting the constant to its justification.

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7. `kb code-ingest` — what it extracts

```
kb code-ingest                          # defaults are usually right
kb code-ingest --db data/app.sqlite     # name a database explicitly
kb code-ingest --json                   # machine-readable report
```

Seven extractors, organised by **artefact kind rather than programming language**, because Strata is adopted across many project lines.

Extractor	Section prefix	Reads
`services`	`svc.<name>`	compose files, Dockerfiles, k8s manifests, solution and workspace files
`deps`	`dep.<ecosystem>`	manifests for Python, JavaScript, Java, .NET, Go, PHP, Rust, Swift, Dart
`commands`	`cmd.<purpose>`	package scripts, Makefiles, tox, shell scripts, CI run steps
`tree`	`struct.tree`	the files Git tracks, so an ignored virtualenv never appears
`schema`	`db.<table>`	SQL migrations, Prisma, Alembic, EF, and explicitly named SQLite files
`integrations`	`int.<name>`	environment-variable **keys** only
`api`	`api.<tag>`	in-repo OpenAPI and Swagger contracts

`tree` always finds something, so the zero-detection rule is: `kb code-ingest` exits `1` when **no extractor other than `tree`** found anything. A document holding only a folder listing is a misconfiguration, not knowledge.

7.1 Two security properties

These are design properties, not conveniences.

1. **Keys, never values.** The only environment file the integrations extractor opens is an `.env.example`, `.sample` or `.template` — never a real `.env` — and its pattern has no capture group around the value at all, so there is nothing to leak whatever a matched line contains. Compose `environment:` blocks *are* read and do hold real inline values, but only the key survives.
2. **Explicit database input.** SQLite schema is read only from paths you name on the command line. A scratch or fixture database in the repository can never become published company knowledge by accident.

7.2 Stated parser limits

These are limits, not bugs.

- `schema` recognises `CREATE TABLE` and `ALTER TABLE … ADD COLUMN` in sorted filename order. It does not implement a SQL dialect.
- A `CREATE TABLE` for a name already created in a *different* migration directory is dropped with a warning naming both files, and an `ADD COLUMN` applies only inside the directory of its `CREATE TABLE`. Two independent schemas sharing a table name are never merged into one that exists nowhere.
- `RENAME`, `DROP`, `ADD CONSTRAINT` and `ADD INDEX` warn as out of scope.
- A table name outside the recognised identifier shape is skipped **with a warning naming the file**. The reader counts `CREATE TABLE` keywords and compares against what it recognised, so a skip is never silent.
- EF Core migrations are recognised by table name only.

- `build.gradle` is matched by regex, not parsed as a DSL.
- Maven `<plugins>`, `setup.cfg` `extras_require`, `pnpm-workspace.yaml` and compose `healthcheck:` are not read.

7.3 A destination it did not write is refused

If `.kb/<doc_id>/` already holds Markdown or a manifest and that manifest is missing, unreadable, not titled `… — code knowledge`, or lists a section id outside the seven generated prefixes, `kb code-ingest` exits 1 and writes nothing. There is no override flag — move the document or choose another `--doc-id`.

7.4 Determinism and the dirty-tree caveat

Re-running on the same commit on the same platform produces byte-identical files, which is what makes the CI publish a true no-op on an unchanged tree.

Two honest limits: a few glob matches case-normalise per operating system, so the guarantee is per-platform (CI is always Linux, so what gets published is stable); and the manifest's revision derives from the **HEAD commit** while your working tree may hold uncommitted changes. `kb code-ingest` detects that and prints a `dirty_tree` warning.

Local `-code` is what CI would produce if your tracked files were committed now; `kb doctor` warns when your installed strata-kb differs from the version `kb-code.yml` pins.

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8. Keeping knowledge current

Neither document needs new per-ticket discipline from you.

8.1 `-code` needs nothing

`-code` is never committed. `.gitignore` carries `.kb/*-code/`, and the document is regenerated from your working tree whenever a command needs it: `kb svc note`, `kb build`, `kb doctor` and `kb publish` refresh it first when it is missing, older than `HEAD`, or your tracked files have changes. A new file counts once it is tracked (`git add`). `kb ticket check` reads the hub copy first — that is what the SA grounded on (§4.1).

`kb doctor`'s published-snapshot comparison leaves `-code` out entirely (dev repos never match the hub on it); `kb publish` is what actually refreshes it and sends it to the hub.

`kb-code.yml` re-runs `kb code-ingest` → `kb build` → `kb ci-publish` on every push to the default branch, so the hub reflects the current commit once its PR merges.

Trigger	Steps	Publishes
push to default branch, or manual dispatch	code-ingest → build → ci-publish	yes
pull request	build (which regenerates `-code` from the merge ref)	**never**

`-code` has one producer, `kb code-ingest`: `kb summarize` refuses it. A hub copy that was summarized before 1.4.0 returns to the deterministic L1/L2 on the next CI publish — a one-time diff, not lost information; L3 is unchanged.

`--scaffold-svc` is deliberately never passed in CI. CI must never create `pending` content, since that would fail its own build step.

8.2 `-svc` accrues at handover

`dev-handover` runs, for every service a ticket touched:

```
kb svc note payments-api --ticket ABC-123 \
 --title "Add refund endpoint" --refs "payments-spec §4.2"
```

That appends one row to `hist.<service>`. It is deterministic and **idempotent**: re-running for the same ticket and service updates the row rather than duplicating it.

You never hand-edit `hist.*`. It is an append-only log written only by `kb svc note`. Those sections are facts — "this ticket touched this service" — not claims, so they need no review and never block a build.

`kb svc note` exits 1 on any of: no resolvable repo id; a missing `-svc` or paired `-code` document; an unknown service (a typo must not invent one); an unreadable manifest; or a manifest/file desync.

8.3 Drift in responsibility text

Responsibility text can drift, and the system **self-reports rather than self-heals**.

When `kb code-ingest --scaffold-svc` refreshes a service's code evidence and that section's summary is already `reviewed`, it compares the old and new rendered evidence as a whole, whitespace-trimmed string — not a semantic diff over specific fields. **Any** change beyond leading and trailing whitespace flags `stale-risk: svc.<name>` in the report.

A real change trips it — a file or table added or removed. So does a change in nothing but *how* the evidence is rendered, with no code changed underneath. It is not an error, and it never touches the summary. It is a flag that a human should look.

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9. Troubleshooting

Symptom	Cause	Fix
`kb code-ingest` exits 1: nothing detected	Only `tree` found anything	Check that your compose file, manifests or CI workflows are where the extractors look; name a `--db` if schema is the point
`kb code-ingest` exits 1: destination refused	A hand-curated document sits at that doc id	Move it, or use `--doc-id`
`kb build` fails in CI on a `-svc` section	A `pending` section was committed	Finish the seed: summarize, review, `kb approve`
`kb svc note`: "unknown service"	A typo, or the new service's files are not tracked yet	Copy the id from `services.md`, `git add` the new files and retry
`kb doctor`: "hub `<repo>`-code lags origin/main"	The last `-code` publish has not merged on the hub, or the last `kb-code.yml` run failed	Open the hub PR named in the Actions job summary; merge it or fix the run
`kb doctor`: "kb-code.yml passes --db"	Ingest configuration lives in the workflow	Move the paths to `code_ingest.db` in `.kb/config.yaml`, then re-run `kb init --kind dev`
`kb doctor`: "`.kb/<repo>-code` is tracked by git"	A repo from before 1.4.0	`git rm -r --cached .kb/<repo>-code` and commit, in one PR
`kb doctor`: "hub lag not judged — origin/HEAD is not set"	`origin/HEAD` was never set (a fresh `git remote add` + `push -u`, before anyone ran `git remote set-head`)	`git remote set-head origin --auto`
`kb publish` hangs, then fails citing the Actions run	Plain `kb publish` on a repo whose CI has no tag trigger	Use `kb publish --pr`
`kb ci-publish` refused	This repo is not in the hub's `federation/registry.yaml`	Ask the hub maintainer to add it
`dirty_tree` warning	Uncommitted changes while the manifest revision comes from HEAD	Harmless locally; CI always runs clean
`pr-lint` fails on a Dependabot PR	The gate never self-skips once required	Expected. Decide whether to keep it required for bot PRs.
`pr-lint` fails: `no-review-verdict`	The PR's `## Testing` section carries no `Blocking:` line	Run A5 (`dev-handover`'s merge-risk review) and record its verdict — even a clean review states `Blocking: No`
`pr-lint` fails: `review-blocking`	The `## Testing` section says `Blocking: Yes`	Fix the BLOCKER findings and re-review; a PR with `Blocking: Yes` never merges
`kb ticket fetch` "is not on the hub"	The BA has not published it yet, or it was left out of the plan (an unticked DoR, or a parent mission that moved on)	Ask the BA to `kb publish` (or to say why it was skipped); or pass the ticket file or paste the body
`kb ticket fetch`: "published by several repos"	Two BA repos publish the same id	Qualify it: `kb ticket fetch <repo>:<id>`
`kb ticket fetch` cannot reach the hub	`hub:` unset or unreachable	Fix `hub:` in `.kb/config.yaml` (or `STRATA_KB_HUB`), or use the path or paste fallback
`kb ticket check` fails with `re-run /sa-ticket-ground`	A fact the ticket cites is gone from `-code`	Stop and hand the ticket back to the BA; the SA re-grounds it
`kb ticket check [warn] … review the diff`	A cited section changed but its cited names still exist	Read the diff; implement unless it changes what the ticket means
A citation resolves `stale` mid-implementation	The hub published after the ticket was written	`kb diff`, then ask the analyst. Do not reinterpret the AC yourself.
MCP server unreachable	`STRATA_KB_HUB_URL` or `STRATA_KB_HTTP_TOKEN` unset or wrong	Re-run `kb mcp-setup` (`/kb-mcp-setup`) — it diagnoses which one, and a bare re-run re-verifies without retyping the token
`kb mcp-setup` fails with "token was rejected" after the hub maintainer gave you a fresh one	A bare re-run reads the old token straight back out of `.env` — the prompt only appears when nothing is on disk yet	`STRATA_KB_HTTP_TOKEN=<new-token> kb mcp-setup`, or delete the `STRATA_KB_HTTP_TOKEN` line from `.env` and re-run

The hub-lag check itself runs a full `code-ingest` in a temporary worktree on every `kb doctor` — expect it to add real time to the command. On Windows, the same per-platform glob-casing difference noted in §7.4 can make an otherwise-unchanged repo compare as `lags`; a lag verdict that disappears on Linux (CI) is that caveat, not real drift.

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10. Command summary

Command	Purpose	Exit
`kb mcp-setup [--hub-url URL] [--no-verify]`	Write the hub's HTTP MCP credentials into `.env` and verify them	`0` ok, `1` no value/probe failed
`kb code-ingest [--db p] [--tags t] [--scaffold-svc] [--json]`	Extract code structure into `-code` (`--db`/`--tags` replace `code_ingest:` for that run)	`0` ok, `1` nothing detected or destination refused
`kb svc note <svc> --ticket <id> --title "…" [--no-refresh]`	Append a row to `hist.<svc>`	`0` ok, `1` unknown service or missing document
`kb build [--strict] [--no-refresh]`	Validate the store	`0` ok, `1` error
`kb doctor [--no-refresh]`	Check KB health (dev repos: refresh `-code` first)	`0` ok, `1` error, `2` stale citation
`kb approve <doc> [--section <id>]`	Flip corrected sections to `reviewed`	`0` ok
`kb publish --pr`	Open a hub PR (the seed's first publish)	`0` ok
`kb ticket fetch <id> [--out p] [--hub u]`	Fetch a ticket from the hub into `docs/impl/<id>/ticket.md`	`0` ok, `1` not on the hub, ambiguous, or unsafe id
`kb ticket check <file> [--hub u]`	The ticket's grounding against the hub's `-code`; a moved revision fails only when a cited fact is gone	`0` PASS, `1` FAIL
`kb mission next --from-hub [--repo id]`	Status of every published story, from the hub alone	`0`
`kb resolve <file>`	Check a ticket's citations	`0` ok, `1` broken, `2` stale
`kb diff <doc> --against <rev>`	What an amendment changed	`0`
`kb query <text>` / `kb get <doc> <section>`	Search and fetch	`0`
`kb plan waves <plan-file> [--json]`	Dependency waves of a `docs/impl/<id>/plan.md`	`0` ok, `1` shared file without a dependency, cycle, unknown task, or missing `Depends on:`
`kb impl tidy [--impl-dir]`	Move an old flat `docs/impl/<id>-*` layout into `docs/impl/<id>/`; idempotent, never overwrites	`0` after a report, `1` when the dir is missing
`kb pr lint <file>`	PR evidence gate	`0` PASS, `1` FAIL

Slash commands: `/dev-implement-ticket`, `/dev-design`, `/dev-plan`, `/dev-execute`, `/dev-handover`, `/dev-code-seed`.

11. Reserved names

`<repo_id>-code` and `<repo_id>-svc` are reserved for this repository's own code knowledge. A domain document must never take either suffix.

Section prefixes are a published contract that analyst tooling and future tooling key on. Each has exactly one owner:

Prefix	Document	Owner
`struct.tree`, `svc.<name>`, `db.<table>`, `dep.<eco>`, `int.<name>`, `api.<tag>`, `cmd.<purpose>`	`-code`	extractor
`svc.<name>`	`-svc`	you (model-drafted)
`flow.<name>`	`-svc`	you
`hist.<name>`	`-svc`	`kb svc note` only

`svc.<name>` collides across the two documents **on purpose**. It is the join key that lets a C4 `Container(alias, label, technology, description)` be filled from `-code` (the first three) and `-svc` (the fourth).